PCK-SIM-001 / SIM-02A Engineering Report

Sensor, Noise and Error Modeling for Candidate Picket Arrays

Program	Ulmo / Picket, PCK-SIM-001
Increment	SIM-02A (sensor, noise and error modeling), authorized by the SIM-01 technical review (ADR-0009)
Run of record	`20261107T214846Z_c56c0048_31da01b`
Code	`picket_sim` 0.1.0, git `31da01b` (clean tree)
Configuration	`configs/sim02a/sim02a_baseline.yaml` (sha256 `c56c00480adbebce...`)
Reproduce	`make sim02a` (equivalently `python -m picket_sim run configs/sim02a/sim02a_baseline.yaml`)
Status	Complete for SIM-02A scope. Submitted for review of numerical results. SIM-02B not begun (gate, ADR-0009). No architecture selected.

All numbers are read from `results/sim02a/20261107T214846Z_c56c0048_31da01b/tables/`; figures are in `results/sim02a/20261107T214846Z_c56c0048_31da01b/figures/`. **Every sensor, ambient, host and channel-error level in this study is an assumed scenario** (72 assumed parameters, 1 published constant, 0 measured values; `tables/parameters_by_provenance.md`). Results that depend on absolute levels are scenario results. Wherever possible results are expressed as ratios (self-noise to ambient, host to ambient) so that they remain valid when measured values replace the assumptions. Nothing here is detection performance, a detection range, or a hardware requirement.

0. Findings

1. **Realized array gain is set by the spatial coherence of the dominant noise, not by geometry alone.** Against diffuse, ambient-limited noise (spherically isotropic, no sensor noise) the band-median realized gain relative to A0 is at most 0.07 dB from 20 to 200 Hz for every candidate up to 1.07 m (1.08 dB for the 4 m envelope V-D), and 0.6 to 5.2 dB from 200 Hz to 2 kHz. When sensor self-noise dominates (self-noise 20 dB above ambient) or the noise is short-correlation (flow-like, L = 0.1 m), the same arrays approach the white-noise gain of their equal weights, 10 log10 N, at every frequency: for example 4.59 dB for A1 and 8.75 dB for P8-E from 20 to 200 Hz in the short-correlation field. Low-frequency array value therefore depends on which noise dominates at the sensors, which only measured sensor, ambient and host data can settle. (High, for the model.)
2. **Under the assumed sensor scenarios, ambient noise dominates almost everywhere, so realized gain follows the diffuse-field values.** The low-noise scenario S1 has band-median self-noise to ambient ratios of -20.0 to -74.4 dB in every band and ambient scenario. The mid scenario S2 is sensor-limited only in the exploratory band B4 under quiet ambient (+4.0 dB). The low-cost scenario S3 is near or above quiet ambient from 200 Hz up (+1.8 dB in B2, +12.1 dB in B3) and cannot acquire B4 at all (32 kHz sampling, 12 kHz declared bandwidth). In S2, 16-bit quantization noise exceeds the analog noise in B3 and B4 (-2.7 and -3.2 dB margins): gain staging, not the hydrophone, sets its noise floor there. (Moderate: assumed scenarios.)
3. **At equal footprint, six and eight elements on a circle widen the conventional band under -3 and -6 dB criteria, but much less under -10 dB.** Guaranteed band ratios over horizon steering at the 0.45 m footprint: 2×2 square 1.28 / 1.14 / 1.03, six-element circle 2.53 / 2.37 / 1.28, eight-element circle 3.78 / 2.80 / 1.32 (-3 / -6 / -10 dB). The circles' intrinsic sidelobes, near -8 dB, cap the -10 dB band. They also add gain where sensors are uncorrelated: 8.58 dB for the eight-element circle against 5.88 dB for the square from 2 to 10 kHz in isotropic noise (band medians). (High, for narrowband DAS.)
4. **Four-element volumetric envelopes have no band free of -6 dB narrowband ambiguity lobes for every horizon steering direction**; their guaranteed -3 dB bands are 1.06 to 1.14. Scaling the envelope to 4 m moves the band down to 146 to 182 Hz and raises low-band gain, at the cost of narrowband ambiguity above that. Broadband and TDOA processing, which may resolve these ambiguities, is SIM-02B. (High for the narrowband statements.)
5. **Calibration and synchronization tolerances scale with frequency, and bearing statistics are a stricter constraint than signal loss.** Signal loss follows the verified closed form; total phase error grows with frequency, so timing and position errors dominate at high frequency. Under the moderate scenario E2 (0.5 dB, 3 deg, 3 mm, 1 us) the mean signal loss is at most 0.17 dB up to 15 kHz, but under E3 (1 dB, 10 deg, 1 cm, 10 us) it reaches 3.2 to 4.5 dB at 15 kHz, and under E4 (100 us timing) the coherent gain is essentially lost above 5 kHz. Bearing statistics are stricter than signal loss: for the 8-element compact circle at 15 kHz, E3 costs 4.4 dB of signal but gives a gross azimuth error (a wrong lobe higher than the true one) in 65 % of trials (95 % interval 62 to 68 %). (High for the model; error levels assumed.)
6. **Line arrays remain structurally ambiguous left/right under any channel errors.** Their scanned response is exactly symmetric in azimuth because the beamformer's steering vectors depend only on the angle to the axis; calibration errors cannot break the tie (D02: unresolved ambiguity identified). (High; mathematical.)
7. **Arrays reject host noise only when the target is away from the host direction and the frequency is high enough.** With a host source at +10 dB relative to ambient, the median realized gain at 100 Hz is 0.0 to 0.2 dB for every candidate up to 1.07 m (2.1 to 3.2 dB for the 4 m envelopes). At 2 kHz the median over target directions reaches 4.7 to 12.0 dB, but the worst steering direction stays at 0.3 to 1.7 dB. Assumed generic host placements; no vessel geometry. (Moderate.)
8. **Pareto results change with the provisional tolerances and are not a basis for elimination.** Doubling the SIM-01 engineering tolerances adds A2 to the dominated set; in the SIM-02A attribute set V-D is dominated by V-DS only at 1x and 2x tolerance. Cost, installation, calibration burden and power are not modeled (D06).

1. Scope and gating

SIM-02A implements the authorized sensor, noise and error modeling: hydrophone sensitivity and self-noise, preamplifier and ADC noise, clipping and dynamic range; white, diffuse, horizontally isotropic, directional and short-correlation noise fields; scenario-based host noise; gain, phase, position and timing errors; Monte Carlo sensitivity with statistical uncertainty. Analyses span 20 Hz to 20 kHz with summaries for the D01 study bands: B1 20-200 Hz, B2 200 Hz-2 kHz, B3 2-10 kHz and B4 10-20 kHz (exploratory, contingent on acquisition). Conventional DAS remains the baseline. Not in SIM-02A: broadband processing, TDOA, detection experiments, false-alarm analysis (SIM-02B), and acoustic vector sensor modeling beyond a feasibility assessment (`docs/avs_feasibility.md`).

2. Candidates

`tables/candidates.md`; `figures/s2a_fig01_candidates.png`. Footprint classes by maximum dimension: C = 0.45 m (the A1 aperture), E = 0.90 m (the A2 aperture), I = A4 envelope, D = 4 m class (assumed).

ID	Family	Footprint	N	Max dim (m)	Geometric ambiguity	Exact grating onset (Hz)	Worst-case -3 / -6 / -10 dB ambiguity onset (Hz)
A0	reference	none	1	0	omnidirectional	n/a	n/a
A1	linear	C	4	0.45	conical	5,000	4,432 / 4,229 / 4,060
A2	linear	E	4	0.90	conical	2,500	2,216 / 2,114 / 2,030
A3	planar	C (0.42 m)	4	0.42	mirror	2,500	1,876 / 1,667 / 1,506

ID	Family	Footprint	N	Max dim (m)	Geometric ambiguity	Exact grating onset (Hz)	Worst-case -3 / -6 / -10 dB ambiguity onset (Hz)
A4	volumetric	I	4	1.07	none	857	596 / 439 / 421
SQ4-C	planar	C	4	0.45	mirror	2,357	1,769 / 1,572 / 1,420
P6-C	planar	C	6	0.45	mirror	3,849	3,252 / 3,043 / 1,640
P8-C	planar	C	8	0.45	mirror	n/a	4,787 / 3,576 / 1,687
SQ4-E	planar	E	4	0.90	mirror	1,179	884 / 786 / 710
P6-E	planar	E	6	0.90	mirror	1,925	1,626 / 1,521 / 820
P8-E	planar	E	8	0.90	mirror	n/a	2,394 / 1,788 / 843
V-C	volumetric	C	4	0.48	none	1,904	1,325 / 975 / 935
V-D	volumetric	D	4	4.28	none	214	149 / 110 / 105
V-DS	volumetric	D	4	4.28	none	413	180 / 140 / 114

A1 and A2 form the linear family L4(d), d = 0.15 and 0.30 m (D05). The intermediate envelope reproduces A4 exactly (self-check). The 8-element circle has no exact grating lobe below 20 kHz (its baselines are not commensurate); its narrowband ambiguity onsets are set by high near-grating lobes.

3. Models and verification

Model reference: `docs/sensor_noise_error_models.md`; decisions ADR-0010 to ADR-0013.

- **Acquisition chain:** element thermal noise 4kT Re(Z) for a lossy capacitance, preamplifier voltage noise (with 1/f corner) and current noise, ideal ADC quantization, clipping at the first stage to saturate; all input-referred to pressure through the element sensitivity.
- **Noise fields:** unit-power spatial coherence models; directional densities integrated by sphere quadrature with a two-grid convergence check; every covariance checked for Hermitian positive semidefiniteness and unit diagonal.
- **Realized array gain (RAG):** narrowband output SNR of uniform DAS divided by the SNR of a single identical channel (A0) in the same noise model; self-noise to ambient ratio eta and host to ambient ratio as parameters.
- **Channel errors:** independent Gaussian gain (dB), phase, 3-D position and timing errors per channel; nominal beamformer; Monte Carlo with 1,000 trials per scenario case and 400 per sweep case, with 95 % intervals on every statistic.

Verification. 166 automated tests pass at the run-of-record commit, including closed-form acquisition noise, tabulated J0 values, coherence models against independent quadrature, RAG limits, Monte Carlo means against closed forms for five error types, statistical interval coverage, structural-ambiguity classification, provenance and Nyquist enforcement, and bitwise reproducibility of the pipeline.

Self-checks embedded in the run of record (`record.json`):

```
* Realized gain in the white field versus 10 log10 N: max |difference| 1.8e-15 dB.
* Realized gain in isotropic noise without sensor noise versus the SIM-01 DI: max |difference| 5.3e-15 dB.
* Monte Carlo mean signal power versus the closed form: 1664 cases, max |z| = 3.33, fraction beyond 3 standard errors 0.0024 (0.0027 expected); no case wid
* -3 dB thresholds versus the SIM-01 run of record: 32 comparisons, max relative difference 2.7e-04.
* A4 envelope versus SIM-01 A4 coordinates: max |difference| 0.0 m.
* Parameters by provenance: 72 assumed, 1 published, 0 measured. Warnings: none.
```

4. Results

4.1 Acquisition chain and dynamic range (D01)

`tables/acquisition_*.md`; `figures/s2a_fig02_self_noise.png`, `figures/s2a_fig03_dynamic_range.png`.

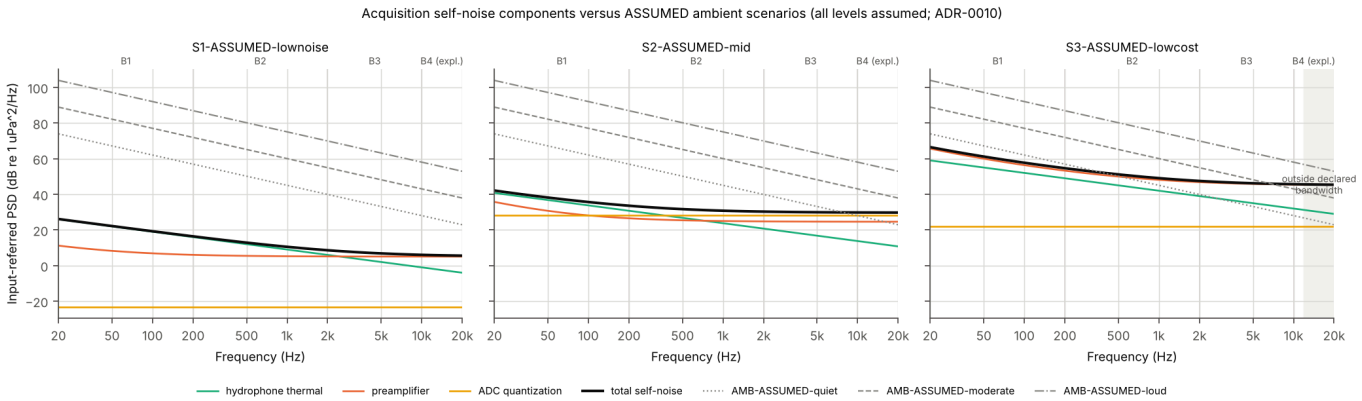


Figure 1. Input-referred self-noise components of the three assumed sensor scenarios against the three assumed ambient scenarios.

Self-noise to ambient ratio eta, band medians (dB), for the moderate and quiet ambient scenarios:

Sensor scenario	Ambient	B1	B2	B3	B4
S1-ASSUMED-lownoise	AMB-ASSUMED-quiet	-44.4	-36.5	-27.2	-20.0
S1-ASSUMED-lownoise	AMB-ASSUMED-moderate	-59.4	-51.5	-42.2	-35.0
S2-ASSUMED-mid	AMB-ASSUMED-quiet	-28.3	-17.3	-4.3	+4.0
S2-ASSUMED-mid	AMB-ASSUMED-moderate	-43.3	-32.3	-19.3	-11.0

Sensor scenario	Ambient	B1	B2	B3	B4
S3-ASSUMED-lowcost	AMB-ASSUMED-quiet	-5.6	+1.8	+12.1	+19.8 (not acquirable)
S3-ASSUMED-lowcost	AMB-ASSUMED-moderate	-20.6	-13.2	-2.9	+4.8 (not acquirable)

- **Ambient-limited operation is the norm in these scenarios.** Only S3 (low-sensitivity element, noisier input stage) approaches or exceeds quiet ambient above 200 Hz. When eta is below about -10 dB, Section 4.2 shows the realized gain is within 0.36 dB of its no-sensor-noise value (all candidates, diffuse fields, all bands).
- **The exploratory band B4 depends on acquisition choices (D01).** S3 cannot acquire it (Nyquist 16 kHz; declared usable band to 12 kHz). The pipeline reports such bands as "not acquirable" rather than extrapolating.
- **Gain staging matters.** In S2 the ADC quantization noise exceeds the analog noise in B3 and B4 (analog minus ADC: -2.7 and -3.2 dB), so more preamplifier gain would lower the noise floor at the cost of clipping headroom. S1 and S3 keep the ADC more than 23 dB below the analog noise.
- **Dynamic range (moderate ambient)** over the total in-band noise, for the broadband class (12.04 dB crest factor): B1 / B2 / B3 / B4: S1 58.4 / 65.4 / 73.2 / 80.5 dB; S2 58.4 / 65.4 / 73.1 / 80.2 dB; S3 50.4 / 57.3 / 63.3 / n/a dB. Transients with a 25 dB crest factor lose a further 13 dB of headroom relative to broadband signals. Clip levels: 173.0 dB re 1 uPa peak (S1, S2) and 165.0 dB (S3), set by the ADC full scale.

4.2 Realized array gain versus noise structure and sensor noise

`tables/rag_*.md`, `tables/rag_band_summary.csv`; `figures/s2a_fig04_rag_isotropic.png`, `figures/s2a_fig05_rag_by_field.png`.

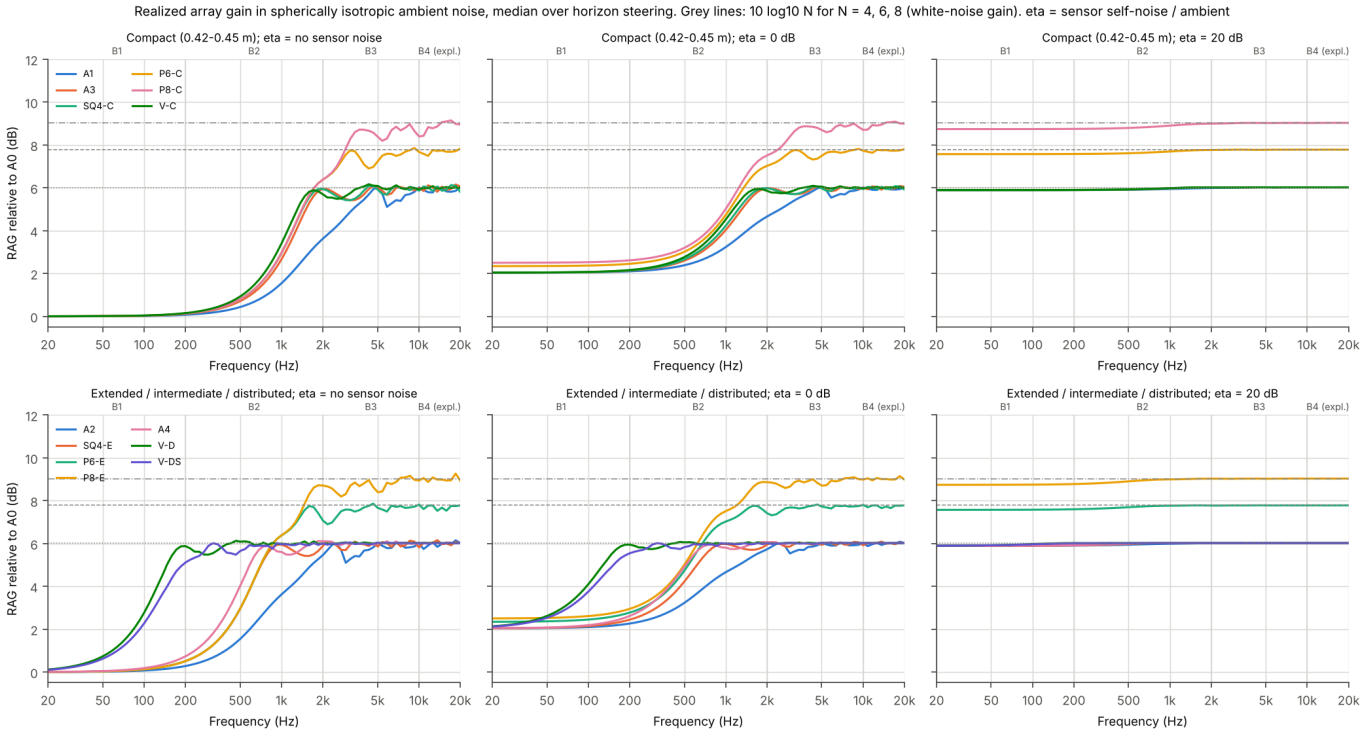


Figure 2. Realized array gain relative to A0 in spherically isotropic ambient noise, median over horizon steering, for no sensor noise, eta = 0 dB and eta = +20 dB.

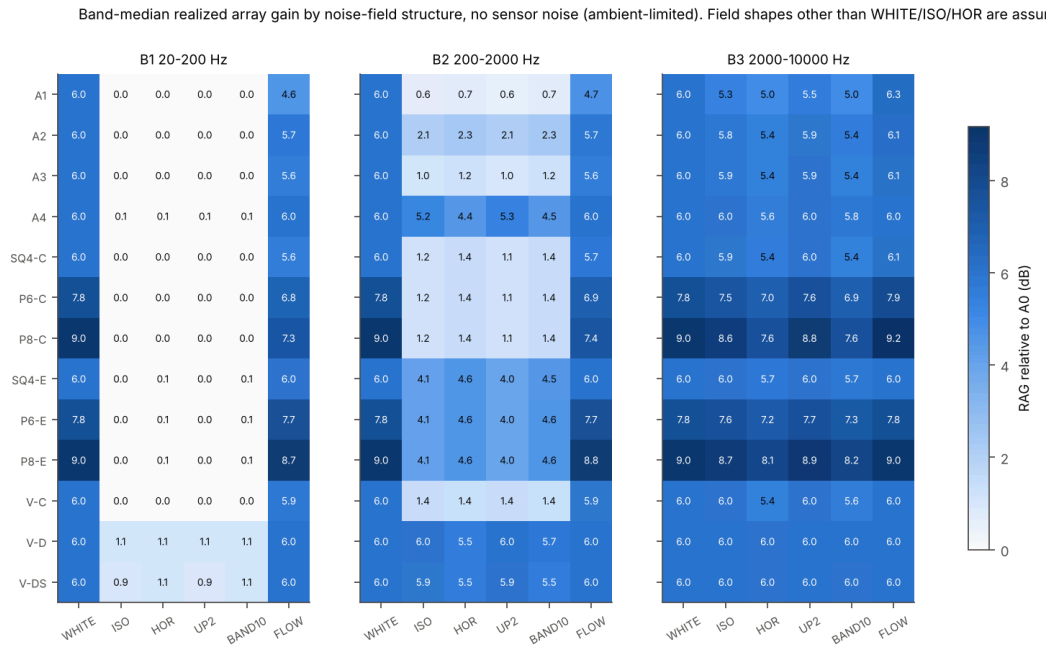


Figure 3. Band-median realized gain by noise-field structure, no sensor noise.

Band-median RAG (dB) with no sensor noise, selected candidates:

ID	ISO B1	ISO B2	ISO B3	HOR B2	UP2 B2	FLOW B1	FLOW B2	WHITE
A1	0.01	0.63	5.27	0.75	0.61	4.59	4.66	6.02
A2	0.03	2.11	5.79	2.34	2.06	5.70	5.75	6.02
A3	0.01	1.04	5.86	1.24	1.00	5.55	5.62	6.02
SQ4-C	0.01	1.17	5.88	1.39	1.12	5.63	5.70	6.02
P6-C	0.01	1.17	7.49	1.39	1.12	6.77	6.87	7.78
P8-C	0.01	1.17	8.58	1.39	1.12	7.32	7.44	9.03
SQ4-E	0.05	4.08	5.96	4.57	3.98	6.01	6.01	6.02
P6-E	0.05	4.09	7.59	4.59	3.99	7.68	7.71	7.78
P8-E	0.05	4.09	8.71	4.59	3.99	8.75	8.80	9.03
A4	0.07	5.20	6.00	4.42	5.34	6.02	6.02	6.02
V-C	0.01	1.36	5.97	1.35	1.37	5.85	5.88	6.02
V-D	1.08	5.97	6.02	5.53	6.02	6.02	6.02	6.02
V-DS	0.91	5.90	6.01	5.53	5.94	6.02	6.02	6.02

- **Diffuse fields (ISO, HOR, upward-weighted, horizontal band) give similar values**: relative to the isotropic field, the directional shape changes band medians by at most 1.0 dB (largest: P8-C, B3, horizontal field; above 1 dB in 1 of 156 comparisons). Spatial coherence over the aperture matters far more than its directional shape.
- **Short-correlation noise behaves like sensor noise**: with L = 0.1 m smaller than most element spacings, it is nearly uncorrelated between sensors and the gain approaches 10 log10 N even at 20 Hz.
- **Sensor noise raises gain relative to A0.** In isotropic ambient, A1's B1 median rises from 0.01 dB (no sensor noise) to 2.05 dB at eta = 0 dB and 5.89 dB at eta = +20 dB. This is not an improvement in absolute sensitivity: the single-channel reference is degraded by the same sensor noise, and the array averages it down.
- The SIM-01 statement that 6.02 dB is the white-noise gain of four equal weights (ADR-0008) is reproduced exactly: RAG in the white field equals 10 log10 N for every candidate (self-check).

4.3 Host-noise scenarios

`tables/host_noise.md`; `figures/s2a_fig06_host_noise.png`. Generic host sources (assumed): near field at 3 m and 15 m (az 180 deg, el +20 and +10 deg) and a far-field source (az 180, el +5 deg). Ambient isotropic, eta = -10 dB, host to ambient ratio +10 dB.

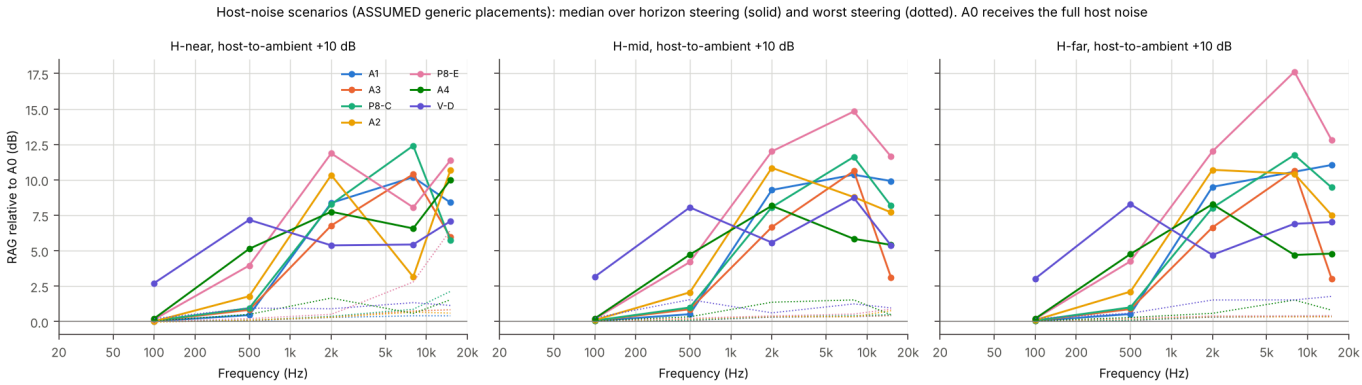


Figure 4. Realized gain relative to A0 with a host source at +10 dB, median (solid) and worst (dotted) over horizon steering.

ID	Host	100 Hz median / worst	2 kHz median / worst	8 kHz median / worst	Toward host azimuth at 2 kHz
A1	H-near	0.0 / 0.0	8.4 / 0.3	10.2 / 0.4	0.3
A1	H-far	0.0 / 0.0	9.5 / 0.3	10.6 / 0.3	0.3
A3	H-near	0.1 / 0.0	6.8 / 0.3	10.4 / 0.7	0.3
A3	H-far	0.1 / 0.0	6.6 / 0.3	10.7 / 0.3	0.3
P8-C	H-near	0.1 / 0.0	8.3 / 0.4	12.4 / 0.9	0.4
P8-C	H-far	0.1 / 0.0	8.0 / 0.3	11.8 / 0.4	0.3
A2	H-near	0.0 / -0.0	10.3 / 0.3	3.2 / 0.7	0.4
A2	H-far	0.1 / 0.0	10.7 / 0.4	10.4 / 0.3	0.4
P8-E	H-near	0.1 / 0.0	11.9 / 0.5	8.1 / 2.8	0.5
P8-E	H-far	0.2 / 0.0	12.0 / 0.4	17.6 / 0.4	0.4
A4	H-near	0.2 / 0.0	7.8 / 1.7	6.6 / 0.6	4.2
A4	H-far	0.2 / 0.0	8.3 / 0.6	4.7 / 1.5	0.6
V-D	H-near	2.7 / 0.2	5.4 / 0.9	5.4 / 1.3	12.8
V-D	H-far	3.0 / 0.2	4.7 / 1.5	6.9 / 1.5	4.1

- A0 receives the full host noise, so arrays that place the host in a sidelobe or null gain more than their no-host value (median RAG above the no-host column).
- When the target azimuth coincides with the host azimuth, compact arrays give at most 1.1 dB up to 2 kHz: DAS cannot separate a target from a host source in nearly the same direction. At 8 and 15 kHz the compact arrays reach up to 5.2 dB, and larger arrays more, because they begin to separate the host source (10 to 20 deg above the horizon) from the horizontal target in elevation.
- At 100 Hz no candidate up to 1.07 m rejects host noise (median RAG 0.0 to 0.2 dB). Low-frequency host noise reaches compact arrays essentially unattenuated by beamforming.

4.4 Channel errors: signal loss, ambiguity and bearing

`tables/mc_scenarios.md`, `tables/mc_sweeps.csv`; `figures/s2a_fig07_mc_verification.png`, `figures/s2a_fig08_mc_ambiguity_bearing.png`. Assumed scenarios (1 sigma, per channel): E1 tight (0.25 dB, 1 deg, 1 mm, 0.1 us), E2 moderate (0.5 dB, 3 deg, 3 mm, 1 us), E3 loose (1 dB, 10 deg, 1 cm, 10 us), E4 unsynchronized (1 dB, 10 deg, 3 cm, 100 us). 1,000 trials per case; noise-free.

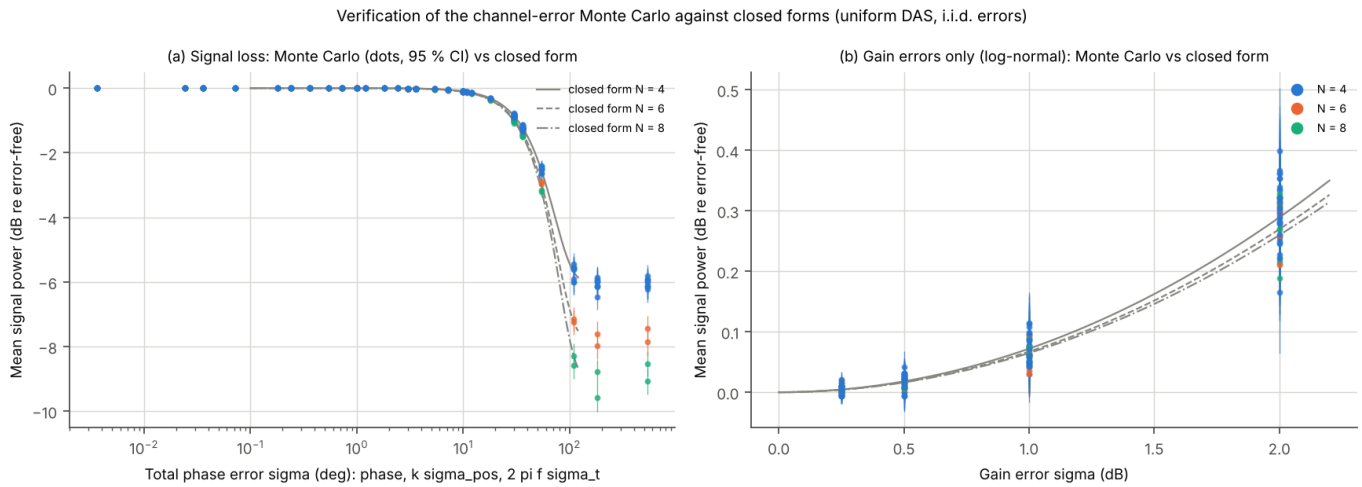


Figure 5a. Mean signal power under channel errors: Monte Carlo (95 % intervals) against the closed form.

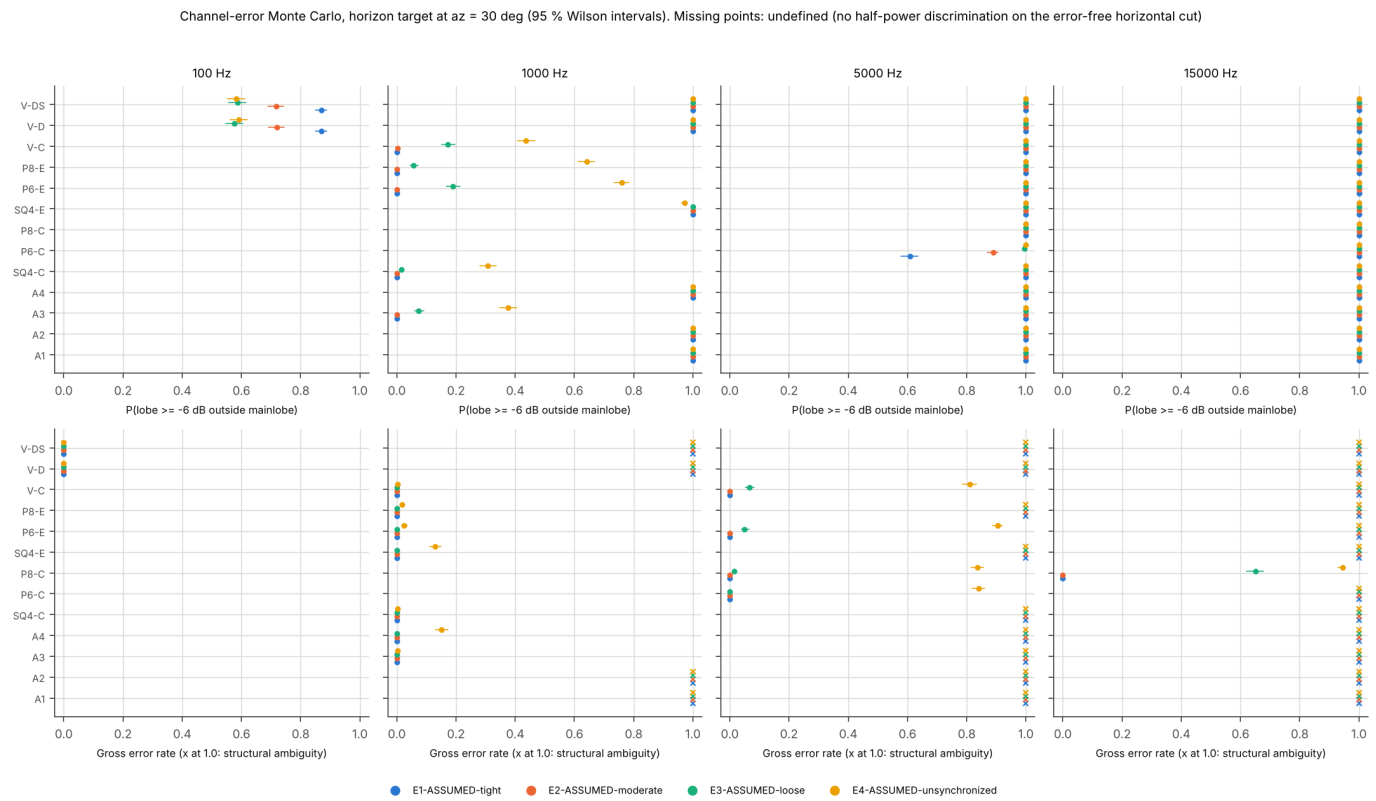


Figure 5. Probability of a lobe at or above -6 dB outside the mainlobe (top) and gross azimuth-error rate (bottom), target at az = 30 deg; x at 1.0 marks structural ambiguity.

Signal loss (mean, all candidates and target azimuths; matches the closed form in all 1,664 Monte Carlo cases):

Scenario	1 kHz	5 kHz	15 kHz
E2 moderate	-0.02 to 0.01 dB	-0.01 to 0.03 dB	0.12 to 0.17 dB
E3 loose	0.00 to 0.08 dB	0.44 to 0.61 dB	3.21 to 4.46 dB
E4 unsynchronized	1.19 to 1.61 dB	5.71 to 9.12 dB	5.51 to 9.09 dB

Under E4 above 5 kHz the loss approaches 10 log10 N (6.02, 7.78, 9.03 dB): the channels add incoherently.

Implied tolerances (derived from the verified closed form, not simulated). Largest single error source that keeps the mean signal loss below 1 dB:

N	Total phase sigma	Timing at 1 / 5 / 10 / 15 kHz	Position at 1 / 5 / 10 / 15 kHz
4	32.4 deg	90 / 18 / 9.0 / 6.0 us	135 / 27 / 13.5 / 9.0 mm
8	29.7 deg	82 / 16.5 / 8.2 / 5.5 us	124 / 25 / 12.4 / 8.2 mm

These are loose bounds. The ambiguity and bearing statistics below fail at much smaller errors.

Ambiguity under channel errors. Errors fill nulls and raise sidelobes toward (1 - exp(-sigma^2))/N. At 1 kHz, the probability of a lobe at or above -6 dB outside the mainlobe rises for the 8-element extended circle from 0.00 (E1, E2) to 0.06 (E3) and 0.64 (E4), and for A3 from 0.00 to 0.07 and 0.38. Under the -10 dB criterion the error-free squares and extended circles already have outside lobes above -10 dB at 1 kHz (A3 -8.5 dB, P8-E -7.8 dB), so calibration cannot help there (the compact circles have no closed half-power mainlobe at 1 kHz).

Noise-free azimuth errors from calibration (target at az = 30 deg; fine error is the RMS error of the correct-side peak):

Candidate, frequency	E1	E2	E3	E4
P8-C, 5 kHz: fine RMS / gross rate	0.17 deg / 0.00	0.54 deg / 0.00	2.53 deg / 0.01	17.4 deg / 0.84
P8-C, 15 kHz	0.14 deg / 0.00	0.47 deg / 0.00	2.26 deg / 0.65	6.00 deg / 0.94
P6-C, 5 kHz	0.19 deg / 0.00	0.63 deg / 0.00	2.92 deg / 0.00	16.8 deg / 0.84
A3, 1 kHz	0.84 deg / 0.00	2.50 deg / 0.00	8.86 deg / 0.00	37.3 deg / 0.00
A1, 1 kHz (structurally ambiguous)	1.40 deg / n/a	4.83 deg / n/a	15.8 deg / n/a	32.7 deg / n/a

- **Broad beams turn small phase errors into large azimuth errors.** At 1 kHz the fine errors of A1, A3, SQ4-C and V-C under E3 are 7.9 to 15.8 deg, about 2 to 2.5 times the same arrays' errors at 5 kHz.
- **Structural ambiguity:** line arrays at every frequency, and other arrays above their (near-)grating onsets (for example A3, SQ4-C and P8-E at 5 kHz, most compact arrays at 15 kHz). There the gross-error rate is undefined, because peak picking cannot resolve the direction whatever the calibration.
- These are calibration-induced biases with no additive noise; their combination with noise-induced errors is a SIM-02B task.

4.5 Ambiguity-criterion sensitivity (D03)

`tables/thresholds_bands.md`, `tables/thresholds_summary.csv`; `figures/s2a_fig09_bands_by_criterion.png`.

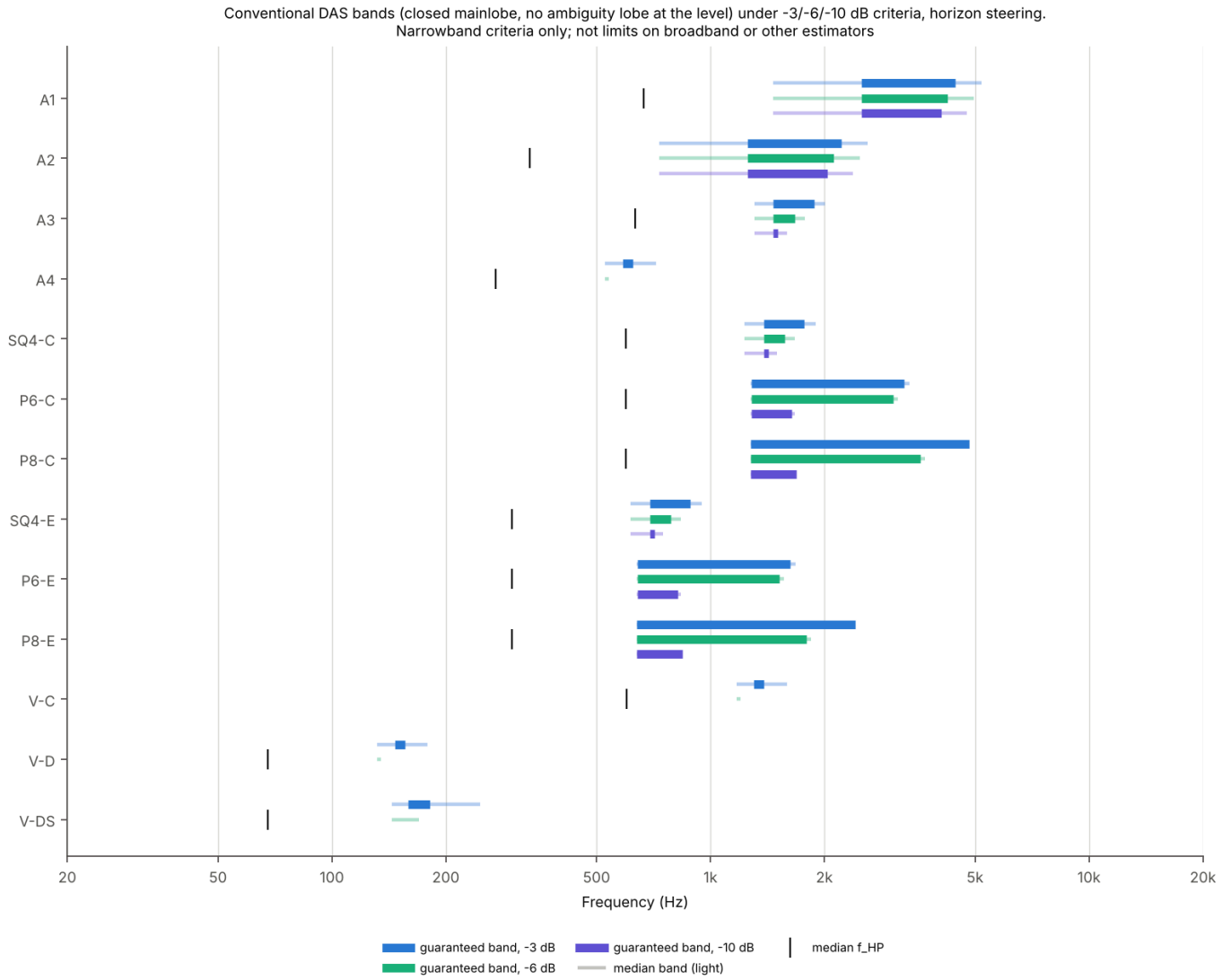


Figure 6. Guaranteed (dark) and median (light) conventional bands under -3, -6 and -10 dB criteria, horizon steering.

ID	f_HP median	Guaranteed -3 dB (ratio)	Guaranteed -6 dB (ratio)	Guaranteed -10 dB (ratio)	Median-band ratios -3 / -6 / -10
A1	667	2,501 to 4,432 (1.77)	2,501 to 4,230 (1.69)	2,501 to 4,063 (1.62)	3.54 / 3.38 / 3.25
A2	334	1,251 to 2,216 (1.77)	1,251 to 2,115 (1.69)	1,251 to 2,031 (1.62)	3.54 / 3.38 / 3.25
A3	633	1,465 to 1,877 (1.28)	1,465 to 1,669 (1.14)	1,465 to 1,506 (1.03)	1.54 / 1.36 / 1.22
A4	270	586 to 623 (1.06)	empty	empty	1.36 / 1.02 / n/a
SQ4-C	597	1,382 to 1,769 (1.28)	1,382 to 1,573 (1.14)	1,382 to 1,420 (1.03)	1.54 / 1.36 / 1.22
P6-C	598	1,283 to 3,253 (2.53)	1,283 to 3,043 (2.37)	1,283 to 1,641 (1.28)	2.63 / 2.46 / 1.31
P8-C	598	1,276 to 4,826 (3.78)	1,276 to 3,578 (2.80)	1,276 to 1,689 (1.32)	3.78 / 2.89 / 1.32
SQ4-E	299	691 to 885 (1.28)	691 to 786 (1.14)	691 to 710 (1.03)	1.54 / 1.36 / 1.22
P6-E	299	642 to 1,626 (2.53)	642 to 1,522 (2.37)	642 to 820 (1.28)	2.63 / 2.46 / 1.31
P8-E	299	638 to 2,413 (3.78)	638 to 1,789 (2.80)	638 to 844 (1.32)	3.78 / 2.89 / 1.32
V-C	601	1,301 to 1,384 (1.06)	empty	empty	1.36 / 1.02 / n/a
V-D	68	146 to 156 (1.06)	empty	empty	1.36 / 1.02 / n/a
V-DS	68	159 to 182 (1.14)	empty	empty	1.71 / 1.18 / n/a

- **The -3 dB results reproduce SIM-01** (32 thresholds, maximum relative difference 2.7e-4).
- **Stricter criteria shrink bands unevenly.** Line arrays keep most of their band (sidelobes of a 4-element line are -11.3 dB below the mainlobe). Squares and circles lose most of it at -10 dB. Volumetric 4-element envelopes have no guaranteed band at -6 or -10 dB.
- These are criteria for narrowband DAS. No threshold is proposed as a requirement (D03).

4.6 Pareto tolerance sensitivity (D06)

`tables/pareto_tolerance_sensitivity.md`, `tables/pareto_sim02a_attributes.csv`; `figures/s2a_fig10_pareto_tolerance.png`.

Attribute set	x0	x0.5	x1	x2	x4
SIM-01 all_candidates	A1 (by A3)	A1 (by A3)	A1 (by A3)	A1 (by A3); A2 (by A3)	A1 (by A3); A2 (by A3; A4)

Attribute set	x0	x0.5	x1	x2	x4
SIM-01 multichannel	none	none	none	none	none
SIM-02A multichannel	none	none	V-D (by V-DS)	V-D (by V-DS)	none

- Dominance depends on the provisional tolerances. It can appear and then disappear as tolerances grow (V-D at 1x and 2x but not at 4x), because wider tolerances turn strict advantages into ties.
- The SIM-02A attribute set includes realized gain at eta = 0 dB, half-power onset, worst-case -3 dB ambiguity onset and the probability of a -6 dB outside lobe under E2 at 5 kHz. It omits cost, installation, calibration burden, power and broadband performance. **No candidate is eliminated.**

5. Traceability to the reviewer decisions

Decision	How SIM-02A addresses it	Where
D01 study bands and signal classes	20 Hz to 20 kHz with B1 to B4 summaries; B4 flagged exploratory and checked against acquisition; tonal, harmonic, broadband and transient classes in the dynamic-range analysis; no bearing assumed for every class	Sections 4.1, 4.2; `tables/acquisition_dynamic_range.*`
D02 detection first, bearing secondary	Gain relative to A0 is the primary figure of merit; bearing statistics reported for horizontal azimuth only, with unresolved ambiguities identified; no maneuver-based disambiguation	Sections 4.2, 4.4
D03 ambiguity criteria	-3 dB kept (regression against SIM-01), -6 and -10 dB added; median and guaranteed bands; no threshold adopted as a requirement	Section 4.5
D04 A4 notional	Parametric envelopes (compact, intermediate = A4 regression, distributed, draft-limited distributed) reported with their dimensions; no vessel geometry	Sections 2, 4.2, 4.5
D05 candidates	A0 to A4 retained; 6- and 8-element circles and 2×2 squares at two footprints; linear family; AVS feasibility assessment	Section 2; `docs/avs_feasibility.md`
D06 tolerances	Tolerances kept and treated as provisional; sensitivity at 0, 0.5, 1, 2, 4x; no elimination	Section 4.6
Provenance	Every parameter tagged; 72 assumed, 1 published, 0 measured	`tables/parameters_by_provenance.md`
Fail visibly	Nyquist, bandwidth, provenance, PSD, convergence and normalization checks raise errors; undefined statistics reported as n/a with reasons	`docs/sensor_noise_error_models.md`

6. Limitations

- **All levels are assumed.** Absolute self-noise, ambient, host and error levels are illustrative. The ratio-based results (gain versus eta, versus host ratio, versus error sigma) transfer; the scenario-specific numbers do not.
- **Narrowband only.** Ambiguity conclusions describe single-frequency patterns; broadband processing is SIM-02B.
- **Noise-free bearing statistics.** Calibration-induced biases only; errors from additive noise need task-level experiments (SIM-02B).
- **Independent Gaussian channel errors.** Correlated errors (temperature drift common to channels, systematic position bias, clock drift) are not modeled.
- **Idealized elements.** Point, omnidirectional elements; no element directivity, host baffling or scattering; no near-field effects on the signal (host sources only).
- **Steering statistics** use the horizon ring (24 directions); Monte Carlo uses 4 target azimuths.
- **The Pareto analysis** covers evaluated attributes only.

7. Recommendations for SIM-02B (based on the verified SIM-02A results)

These recommendations need the reviewer's decision before SIM-02B begins (ADR-0009).

1. **Detection experiments against A0 in both noise regimes.** Finding 1 shows the array's value reverses between ambient-limited diffuse noise and sensor- or flow-limited noise. SIM-02B detection experiments at fixed false-alarm probability should span eta from about -40 to +20 dB and include the short-correlation field, for tonal, harmonic, broadband and transient classes.
2. **Broadband beamforming and TDOA for the candidates with narrowband ambiguity.** Priority: the volumetric envelopes (A4, V-C, V-D, V-DS), the 2×2 squares above their grating onsets, and the circles under the -10 dB criterion. Measure task-level ambiguity (probability of a wrong-lobe estimate) and bearing error statistics rather than sidelobe levels (D03).
3. **Carry channel errors into task-level metrics,** with the E1 to E4 scenarios and an explicit synchronization model for distributed modules. Timing error is the dominant term at high frequency and directly limits TDOA.
4. **Replace assumed inputs with published or measured values before any result informs requirements:** candidate hydrophone and preamplifier noise data, published ambient-noise models with provenance, host noise and vibration measurements, and flow-noise correlation at host speeds.
5. **Candidate set for SIM-02B:** A0 as the baseline; the linear family (bearing with known left/right ambiguity); the 6- and 8-element circles at both footprints; A4 and the distributed envelopes for broadband tests. The 2×2 squares remain as references. Feasibility data collection for acoustic vector sensors in parallel (`docs/avs_feasibility.md`).
6. **Correlated-error and drift models** if the reviewer considers long-duration calibration stability relevant to proliferated hosts.

8. Reproducibility and provenance

- `make sim02a` runs the test suite and the experiment into a new immutable directory. `record.json` holds the resolved configuration and hash, git commit, software versions, timings, self-checks and SHA-256 hashes of every output. `tables/parameters_by_provenance.*` lists every parameter's provenance.
- `results/sim02a/RUN_INDEX.csv` lists all SIM-02A runs. Two earlier runs are retained unchanged with errata (`results/sim02a/ERRATA.md`). Both contained defects in the Monte Carlo bearing statistics only (a tie-breaking artifact for line arrays, then a tie test that absorbed gross errors). Every other table is byte-identical across all three runs (29 of 32 tables identical; the 3 Monte Carlo tables differ as intended).